



METFORMIN TABLET 500MG

Each tablet contains:-
Metformin HCl 500mg

Pharmacology:

Metformin is a biguanide with antihyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.

Metformin may act via 3 mechanisms: Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis; in muscle by increasing insulin sensitivity, improving peripheral glucose uptake and utilisation; and delay of intestinal glucose absorption.

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase. Metformin increases the transport capacity of all types of membrane glucose transporters (GLUT). In humans, independently of its action on glycaemia, metformin has favourable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies; metformin reduces total cholesterol, LDL cholesterol and triglyceride levels.

Pharmacokinetics:

After an oral dose of metformin, T_{max} is reached in 2.5 hours. Absolute bioavailability of a 500mg or 850mg metformin tablet is approximately 50-60% in healthy subjects.

Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Indications:

It is used in the treatment of maturity on-set diabetes mellitus not responsive to diet alone or diet plus treatment with a sulphonylurea.

Side effects / Adverse reactions:

Anorexia, nausea, vomiting and diarrhoea, are not uncommon; occasionally a metallic taste, loss of weight, weakness, lassitude or skin rashes may occur.

Lactic acidosis possibly heralded by vomiting, abdominal pain, hyperventilation and diminished consciousness may occur.

Contraindications:

Metformin hydrochloride should not be given to patients with impaired hepatic or renal function, cardiovascular collapse, congestive heart failure, acute myocardial infarction, or other conditions leading to hypotension or to hypoxaemia. It should not be given to patients with diabetes mellitus complicated by acidosis, infection or gangrene or during pregnancy or during surgery.

Severely reduced kidney function (GFR <30 mL/min)

Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

Warnings and Precautions:

Lactic acidosis:

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function:

GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section Contraindications].

Administration of Iodinated contrast agent:

As the intravascular administration of iodinated contrast materials in radiologic studies can lead to renal failure, metformin should be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Surgery:

Metformin hydrochloride should be discontinued 48 hours before elective surgery with general anaesthesia and should not be usually resumed earlier than 48 hours afterwards.

Children and adolescents:

The diagnosis of type 2 diabetes mellitus should be confirmed before treatment with metformin is initiated.

A careful follow-up of the effect of metformin on growth and puberty in metformin-treated children, especially pre-pubescent children, is recommended. Although metformin efficacy and safety in children below 12 did not differ from efficacy and safety in older children, particular caution is recommended when prescribing to children aged between 10 and 12 years.

Other precautions:

All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.

The usual laboratory tests for diabetes monitoring should be performed regularly.

Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or sulphonylureas.

Usage during pregnancy and lactation:

To date, no relevant epidemiological data are available. Animal studies do not indicate harmful effects with respect to pregnancy, embryonal or foetal development, parturition or postnatal development.

When the patient plans to become pregnant and during pregnancy, diabetes should not be treated with metformin but insulin should be used to maintain blood glucose levels as close to normal as possible in order to lower the risk of foetal malformations associated with abnormal blood glucose levels.

Metformin is excreted into milk in lactating rats. Similar data are not available in humans and a decision should be made whether to discontinue nursing or to discontinue metformin, taking into account the importance of the drug to the mother.

Recommended Dosage:

Oral administration:

The initial dose is 500mg thrice daily or 850mg twice daily with meals, gradually increased if necessary to a maximum of 3g daily.

Renal impairment:

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional considerations
60-80	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated.

Interactions with Other Medicaments:

It is reported that metformin reduces the absorption of vitamin B12. The serum level of vitamin B12 is not affected.

Tetracycline may precipitate metformin induced lactic acidosis.

Concomitant use not recommended.

Alcohol:

Increased risk of lactic acidosis in acute alcohol intoxication, particularly in case of fasting or malnutrition and hepatic insufficiency.

Avoid consumption of alcohol and alcohol-containing medications.

Iodinated contrast agents:

See Warnings and Precautions.

Combinations requiring precautions for use:

Glucocorticoids (systemic and local routes), beta-2-agonists, and diuretics have intrinsic hyperglycaemic activity whereas ACE-inhibitors may decrease the blood glucose levels. If necessary, adjust the dosage of the antidiabetic drug during therapy with the other drug and upon its discontinuation.

Symptoms and Treatment for overdosage and antidote(s):

The symptoms are nausea and vomiting, anorexia, weight loss, low epigastric pain, unconsciousness and pulmonary oedema.

In the treatment of lactic acidosis, sodium bicarbonate is given promptly by infusion to correct the acidosis, it may be necessary to give diuretics to avoid sodium overload. Peritoneal dialysis or haemodialysis have been reported to be successful.

Pack size: Blister pack: A box of 10 x 10, 50 x 10 and 100 x 10 tablets per strip.

Pack size (export/tender only): A bottle of 1000 tablets.

Storage conditions: Store at or below 30°C.

Shelf-life: 5 years.

Description: Round, standard convex, white, film-coated tablet with single score on one side only.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.

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Imported by:

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TL643C-R5
Revision Date: 19/07/2018